Beam-induced background rejection with hit timing and direction in the tracker of a muon collider, and the resolutions it requires

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Abstract: At a 10 TeV muon collider the decays of the circulating muons fill the tracker with millions of background hits per bunch crossing. We study how far this beam-induced background can be reduced hit by hit, before any pattern recognition, if every hit carries a time and a direction measurement, and what that implies for track reconstruction in the inner and outer trackers. Three cuts test whether a hit is compatible with a particle coming from the interaction point: on its time corrected for the time of flight, on the extrapolation of its direction to the beam axis, and on the transverse momentum implied by its direction. With a position resolution of 100 μm, a time resolution of 100 ps and a direction resolution of 1.5°, and cuts set to keep 98% of the signal hits each, the background in the inner and outer trackers is reduced by a factor of 57 while 97% of the hits of signal muons and 98% of the tracks are kept. The algorithm-independent lower limit on combinatorial fake tracks of a companion paper then gives 1.9×10^{-2} fakes per bunch crossing: 1.7×10^{-2} in the barrel, 1.7×10^{-3} in the region where tracks cross both barrel layers and endcap disks, and 1.5×10^{-6} in the endcaps. One fake per crossing would be reached at 2.2 times the simulated background density. Scans of the three resolutions show how this margin depends on each of them and set targets for detector R&D.

Keywords: Pattern recognition, cluster finding, calibration and fitting methods; Performance of High Energy Physics Detectors; Timing detectors; Particle tracking detectors

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1 Introduction

A multi-TeV muon collider would combine the discovery reach of a hadron collider of much higher energy with the clean final states of lepton collisions [1, 2]. The price, for the detector, is the beam-induced background (BIB): the decay products of the circulating muons interact with the machine elements near the interaction region and reach the detector as a diffuse flux of low-energy photons, electrons, neutrons and charged hadrons, out of time and out of direction with respect to the collisions [3, 4, 5]. In the tracker this flux produces millions of hits per bunch crossing, with densities above one hit per mm^2 in the innermost layers, and the reconstruction of charged tracks among them is the central challenge of the tracker design.

Several handles against the BIB are known. The hits arrive spread over a few nanoseconds while those of particles from the collision arrive within tens of picoseconds of the expected time of flight, so a timing window removes much of the background [3, 5]. The BIB particles also cross the sensors at directions unrelated to the interaction point, which double-layer sensors or cluster shapes can exploit. Full reconstruction chains built on these ideas, and the detector concepts proposed for a 10 TeV collider [6, 7], show that tracking is possible, with rates of fake tracks that depend on the algorithm used.

This paper asks a complementary question. If every hit carries a time and a direction, how much of the BIB can be removed hit by hit, before any pattern recognition, while keeping nearly all the hits of particles from the interaction point; and does what remains still allow tracks to be found by *any* algorithm? The first part is a selection on three per-hit observables, each testing the compatibility of the hit with a particle produced at the interaction point. The second uses the algorithm-independent lower limit on combinatorial fake tracks derived in a companion paper [8]: the expected number of combinations of background hits that pass the same goodness-of-fit cut as real tracks, which no algorithm keeping the efficiency for real tracks can reject. Both are evaluated for the inner and outer trackers, where the limit is computed from the measured hit densities, layer by layer, in three regions: the barrel, the endcaps, and the intermediate region in which tracks cross both barrel layers and endcap disks. Finally, scanning the position, time and direction resolutions shows how much margin the selection leaves and which resolution matters most. The direction measurement in particular is treated as a capability to be developed: the scans give the precision it would need.

Section 2 describes the simulated samples and the detector, section 3 the per-hit observables and section 4 the selection and its performance. Section 5 applies the fake-track limit to the three regions, section 6 presents the resolution scans and section 7 discusses the limitations.

2 Simulated samples and detector

Detector. We use the MUSIC detector concept for a 10 TeV muon collider [6] [confirm the geometry version and the reference], with a 5 T solenoidal field. Its tracker consists of a vertex detector (VXD), an inner tracker (IT) and an outer tracker (OT), each with a barrel and two endcaps. This paper concerns the IT and OT. Their barrels have three IT layers at radii of 165, 355 and 555 mm and three OT layers at 820, 1154 and 1491 mm; each endcap has seven IT disks between $|z| = 606$ and 2192 mm and four OT disks between 1408 and 2192 mm (table 1). The vertex detector is not used: its hits do not count towards the track-finding criterion and its layers are not included in the fake-track limit. The sensitive areas are taken from the geometry description of the modules.

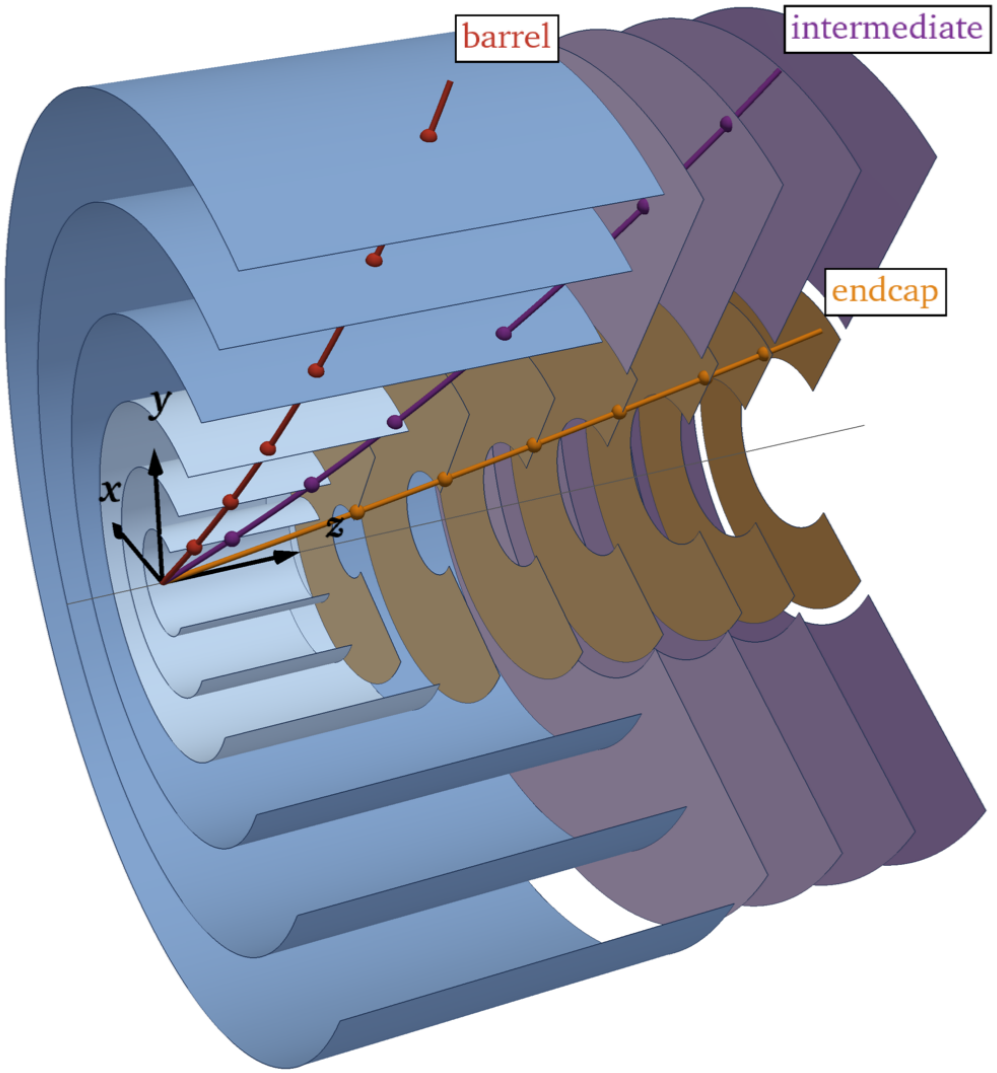


Figure 1: The inner and outer trackers of the MUSIC detector, for $z \geq 0$, with the quarter facing the viewer cut away: barrel layers in blue (IT light, OT dark), IT disks in tan and OT disks in lavender. Three tracks from the interaction point, with their hits, illustrate the three regions of section 5: barrel ($\vartheta = 65°$), intermediate (40°) and endcap (15°). They are drawn with $p_T = 4$ GeV/c so that the bending in the 5 T field is visible.

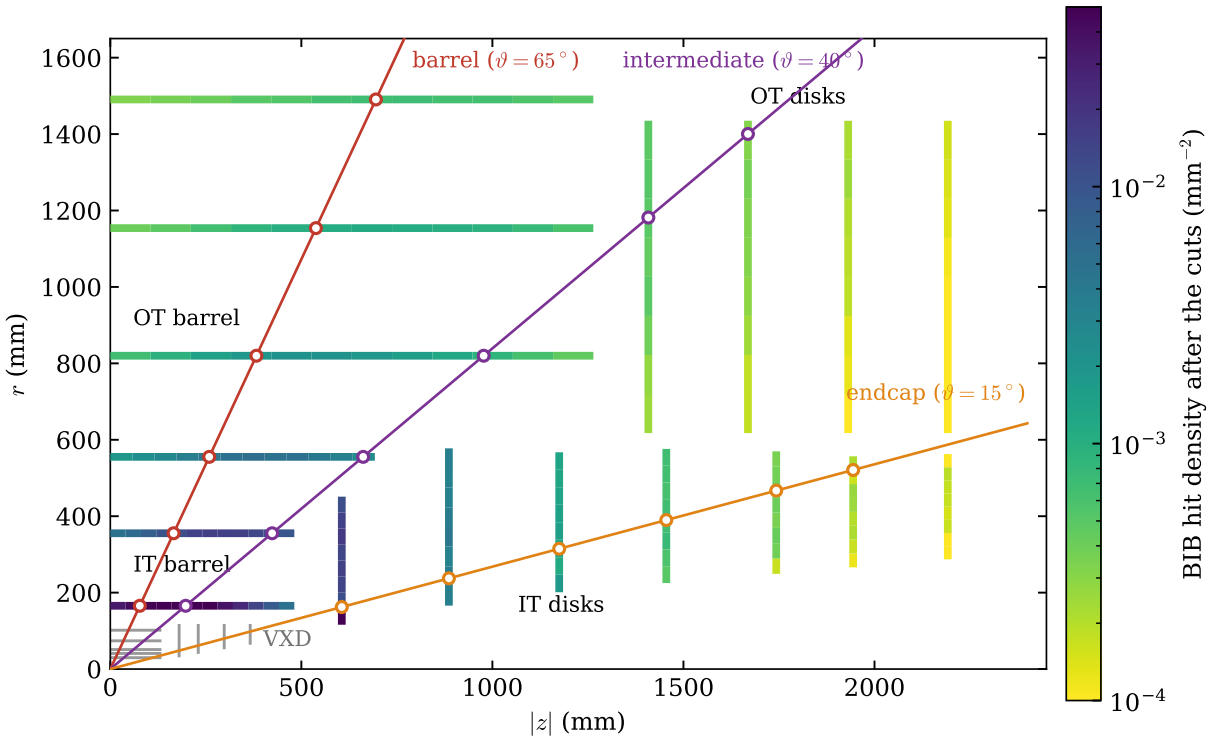


Figure 2: Quarter view of the inner and outer trackers in $(|z|, r)$, to scale: the six barrel layers and the eleven disks of each endcap, coloured by the BIB hit density after the selection of section 4 (as a function of $|z|$ on the barrel layers and of r on the disks, both ends together); the vertex detector, not used here, in grey. The three straight tracks from the interaction point, with their hits, are examples of the three regions of section 5.

Background sample. One bunch crossing of BIB, the sum of the samples of the μ^+ and μ^- beams and of the incoherent pair production at the interaction point, [provenance: generation with FLUKA, detector simulation; who produced it and which version], gives 17.3 million hits in the tracker, of which 10.4 million are in the IT and OT (7.7 million in the six barrel layers). Each hit has its position, its local sensor coordinates, its time and the momentum of the particle that produced it, from which its direction is computed.

Signal sample. 100 000 single muons, with transverse momenta from 1.5 GeV/c upwards, polar angles between 10° and 170°, full azimuth and a longitudinal vertex spread of 1.5 mm. Only the hits of the muon itself are used; those of secondary particles (delta rays) are left out. For the efficiency for high-momentum tracks we extrapolate to $p_T \to \infty$ (appendix A).

Resolutions. The simulated hits are exact. We add Gaussian smearing to represent the detector: $\sigma = 100$ μm on both local coordinates, $\sigma_t = 100$ ps on the time and $\sigma_\alpha = 1.5°$ on each of the two angles that define the hit direction relative to the sensor. The same smearing is applied to signal and background. The position and time values are conventional for present-day pixel detectors with timing; a per-hit direction measurement of 1.5° does not exist today in a single sensor and is used here as a working hypothesis, whose consequences section 6 quantifies.

Table 1: The barrel layers and the endcap disks of the IT and OT: position, extent, sensitive area (for the disks, of both endcaps together), and the mean BIB hit density before and after the selection of section 4.

layer	position (mm)	extent (mm)	area (m^2)	density (mm^{-2})		rejection
				before	after	
IT barrel L0	$R = 165$	$\lvert z\rvert < 482$	1.04	1.58	3.9×10^{-2}	40
IT barrel L1	$R = 355$	$\lvert z\rvert < 482$	2.20	0.552	1.1×10^{-2}	49
IT barrel L2	$R = 555$	$\lvert z\rvert < 692$	5.17	0.244	3.7×10^{-3}	66
OT barrel L0	$R = 820$	$\lvert z\rvert < 1264$	14.0	0.101	1.3×10^{-3}	77
OT barrel L1	$R = 1154$	$\lvert z\rvert < 1264$	19.5	0.0603	8.4×10^{-4}	72
OT barrel L2	$R = 1491$	$\lvert z\rvert < 1264$	28.6	0.0356	5.4×10^{-4}	66
IT disk D0	$\lvert z\rvert = 606$	$116 < r < 451$	1.33	0.273	1.7×10^{-2}	16
IT disk D1	$\lvert z\rvert = 886$	$166 < r < 578$	2.12	0.153	3.6×10^{-3}	43
IT disk D2	$\lvert z\rvert = 1175$	$201 < r < 567$	2.02	0.106	1.4×10^{-3}	77
IT disk D3	$\lvert z\rvert = 1455$	$225 < r < 576$	1.98	0.0774	6.5×10^{-4}	119
IT disk D4	$\lvert z\rvert = 1743$	$249 < r < 570$	1.86	0.0588	3.7×10^{-4}	161
IT disk D5	$\lvert z\rvert = 1944$	$266 < r < 557$	1.72	0.0486	2.0×10^{-4}	239
IT disk D6	$\lvert z\rvert = 2192$	$287 < r < 563$	1.66	0.0425	1.4×10^{-4}	303
OT disk D0	$\lvert z\rvert = 1408$	$618 < r < 1435$	13.9	0.0257	4.4×10^{-4}	59
OT disk D1	$\lvert z\rvert = 1669$	$618 < r < 1435$	13.9	0.0276	2.9×10^{-4}	96
OT disk D2	$\lvert z\rvert = 1931$	$618 < r < 1435$	13.9	0.0231	1.9×10^{-4}	123
OT disk D3	$\lvert z\rvert = 2192$	$618 < r < 1435$	13.9	0.0228	1.1×10^{-4}	202

3 Per-hit observables

A particle from the interaction point reaches each layer at a predictable time and with a direction that points back to the beam line. Three observables, computed for each hit from its position, time and direction alone, measure how far the hit is from this hypothesis.

Longitudinal intercept z_0. In the meridian plane containing the beam axis and the hit, the hit direction defines a straight line through the hit. Its intersection with the beam axis is

$$z_0 = z - r\,\frac{p_z}{p_r}, \tag{1}$$

where r and z are the hit coordinates and p_r, p_z the radial and longitudinal components of the direction. In a solenoidal field the trajectory projected on the meridian plane is a straight line, so a particle from the interaction point has z_0 within the luminous region, smeared by the direction resolution.

Transverse momentum from the direction. In the transverse plane, the circle through the origin and the hit, tangent at the hit to the transverse direction, has curvature

$$\frac{1}{R} = \frac{2\sin\psi}{r}, \tag{2}$$

where ψ is the angle between the transverse direction and the radial direction. It is the curvature a particle from the beam line would need to reach the hit in that direction, and gives the transverse momentum $p_T = 0.3\,B\,R$ (GeV/c, T, m). The cut is on $|1/R|$, which is finite and continuous at infinite momentum.

Corrected time t_{corr}. The time of flight of a muon from the interaction point to the hit along that helix, $s/(\beta c)$, is subtracted from the hit time. The path length s is the arc of the circle above divided by $\sin\vartheta$, the polar angle taken from the hit direction, and the speed β follows from the momentum estimated from the curvature and the muon mass. For a muon from the collision t_{corr} is zero within the time resolution; for a background hit it carries the delay of the background particle.

Distributions. Figure 3 (section 4) shows the three observables in the four parts of the IT and OT for background and signal hits, each after the cuts on the other two observables. The signal peaks sharply: t_{corr} within the time resolution, z_0 within a few centimetres in the IT and about a decimetre in the OT, where the lever arm of the direction measurement is longer, and the curvature within the range of the muon momenta. The background is broad in all three. Its corrected time peaks about 0.3–0.5 ns late, with a long tail of delayed hits and a smaller population arriving early. [the origin of the late peak is not yet understood; Monte Carlo truth may explain it] In the endcaps the background time distribution is broader, and the z_0 of the background is much closer to that of the signal: on a disk the direction measures the polar angle of a track crossing at small angle to the beam, with a short lever arm towards the axis, and in the OT endcap the two z_0 distributions nearly coincide.

4 Selection

Construction of the cuts. Each subsystem (VXD, IT and OT, barrel and endcaps separately) has a symmetric window on each of the first two observables, $|t_{\text{corr}}| \le T$ and $|z_0| \le Z$, and the OT has in addition a minimum transverse momentum, $p_T \ge 2$ GeV/c, applied as a maximum curvature $|1/R| \le 0.3B/p_{T,\text{min}}$. The momentum cut is chosen by hand; T and Z are derived from the signal sample as follows.

1. The reference population is the hits of the signal muons, over the whole momentum spectrum of the sample ($p_T \ge 1.5$ GeV/c), that pass the momentum cut of their subsystem (in the IT, all of them).

2. Each window is set on the *N−1 distribution* of its own observable: the distribution of that observable for the reference hits that pass the *other* cut. Z is the value of $|z_0|$ below which 98% of these hits lie, and the same for T with $|t_{\mathrm{corr}}|$.

3. The two windows depend on each other, so they are found by iteration. We start with Z computed on the reference hits with no time cut, then compute T on the hits passing that Z, then Z again on the hits passing T, and so on, until neither window changes by more than 0.1%. Two or three steps are enough.

Figure 3 shows the N−1 distributions for the four parts of the IT and OT, with the resulting windows. The windows are fixed by the signal alone, not tuned on the background, so they follow automatically when a resolution changes: every point of the resolution scans of section 6 has its own windows, derived in the same way. Since the reference population includes low-momentum muons, whose distributions are wider because of multiple scattering, the fraction of hits kept for high-momentum muons is slightly higher than 98% for each cut alone. Table 2 lists the cuts.

Table 2: Cuts, background rejection factor and signal hit efficiency in the four parts of the IT and OT. Rejection factor: hits before over hits after the cuts, for each cut alone and (in parentheses) on top of the other two. Efficiency: fraction of the muon's own hits kept, extrapolated to $p_T \to \infty$, for each cut alone.

		IT barrel	OT barrel	IT endcaps	OT endcaps
$\lvert t_{\mathrm{corr}}\rvert$	cut	0.233 ns	0.245 ns	0.242 ns	0.241 ns
	rejection	10.3 (4.6)	10.2 (5.8)	16.0 (8.4)	25.4 (4.9)
	efficiency	98.4%	98.7%	98.2%	98.4%
$\lvert z_0\rvert$	cut	52 mm	137 mm	474 mm	349 mm
	rejection	10.5 (4.7)	7.8 (3.6)	4.5 (2.3)	9.0 (1.1)
	efficiency	98.4%	99.4%	97.8%	98.5%
p_T	cut	—	2 GeV/c	—	2 GeV/c
	rejection	—	1.8 (1.8)	—	3.4 (1.5)
	efficiency	—	100%	—	99.9%
all	rejection	48.6	72.0	37.4	96.6
	efficiency	96.9%	98.1%	96.1%	96.9%

Background rejection. Together the cuts reduce the background in the IT and OT from 10.4 million to 185 000 hits per bunch crossing, a factor of 57, with factors between 37 and 97 in the four parts. In the barrel the time and z_0 cuts are comparably powerful, a factor of about ten each alone and four to six on top of the others; the momentum cut adds a factor of two in the OT. In the endcaps the time cut does most of the work, while the z_0 cut adds little on top of the others, only 1.1 in the OT endcaps. The densities after the cuts range from 3.9×10^{-2} mm^{-2} in the innermost IT layer to about 10^{-4} mm^{-2} on the outermost disks; the rejection grows with the distance from the interaction point, from 16 on the first IT disk to 300 on the last (table 1, figure 4).

Signal efficiency. The cuts keep 96–98% of the hits of high-momentum muons in the four parts, 96.8% overall. A track is counted as found if at least five of its hits in the IT and OT survive (vertex-detector hits are not counted); the track-finding efficiency for $p_T \to \infty$ is 97.9 ± 0.2% (figure 5), and 98.1 ± 0.3% for muons confined to the barrel. It falls below about 2.5 GeV/c, where the momentum cut of the OT and the curvature of the track remove hits.

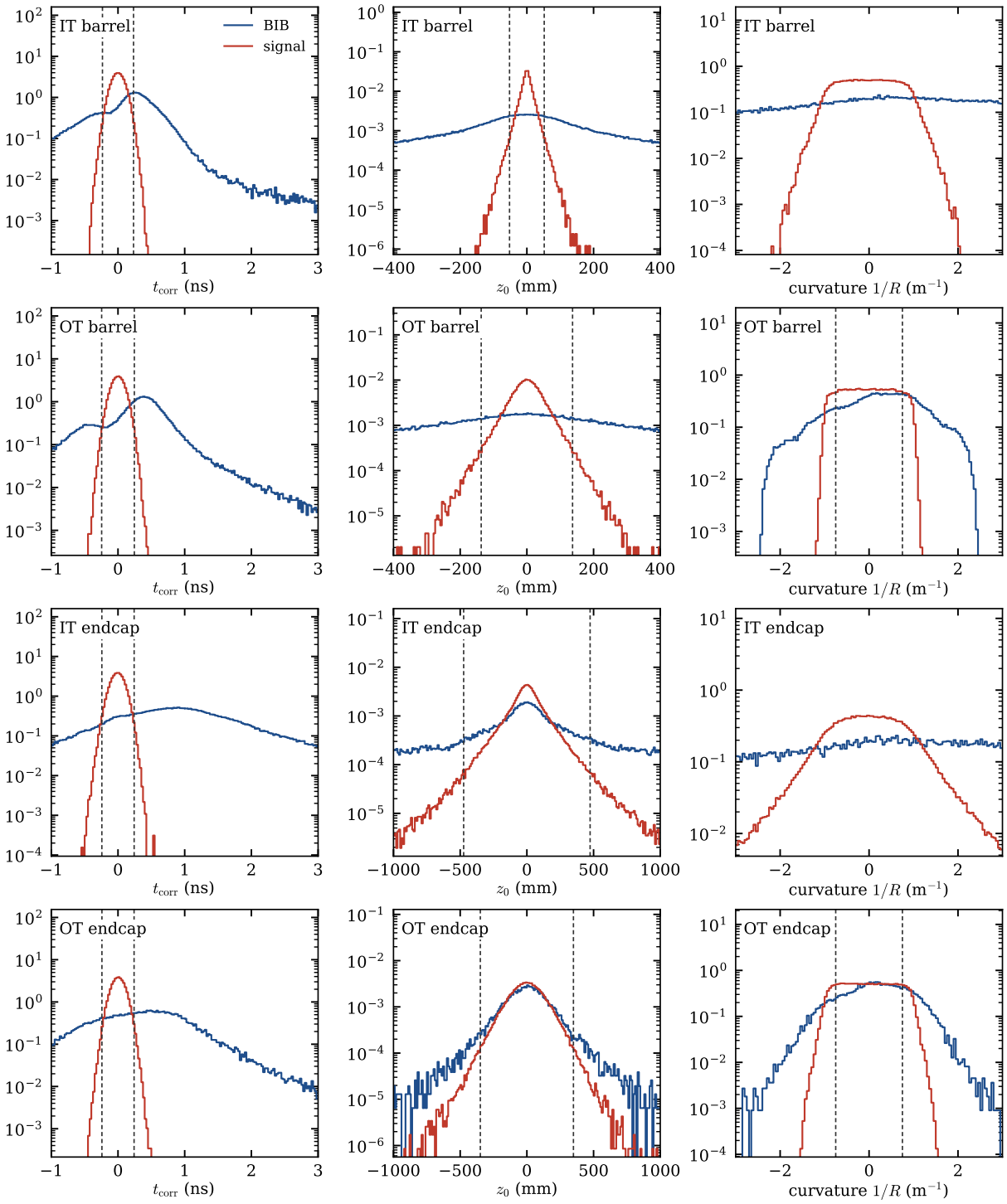


Figure 3: N−1 distributions of the corrected time (left), longitudinal intercept (centre) and curvature (right) in the IT barrel, OT barrel, IT endcaps and OT endcaps (top to bottom). Each panel shows the hits that pass the other cuts of their subsystem, for one bunch crossing of background (blue) and for the signal muons (red); the dashed lines are the cuts of table 2, each set to keep 98% of the signal hits in its own distribution (no curvature cut in the IT). Both curves are normalized to unit area, so that the plot compares the *shapes*: in absolute numbers the signal is many orders of magnitude below the background, one muon leaving one or two hits per subsystem against millions of background hits per bunch crossing.

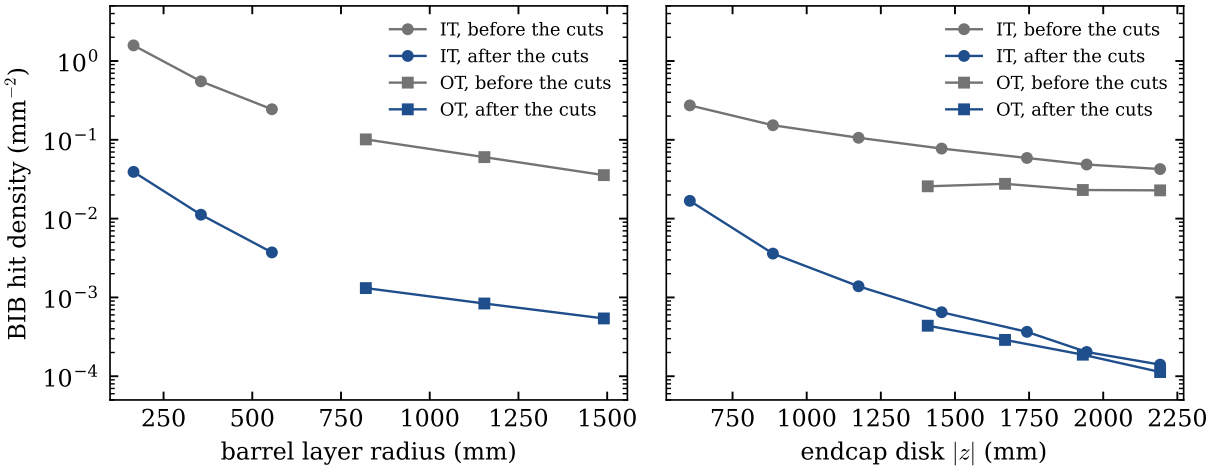


Figure 4: Mean BIB hit density in the barrel layers (left) and endcap disks (right) of the IT (circles) and OT (squares), before and after the cuts.

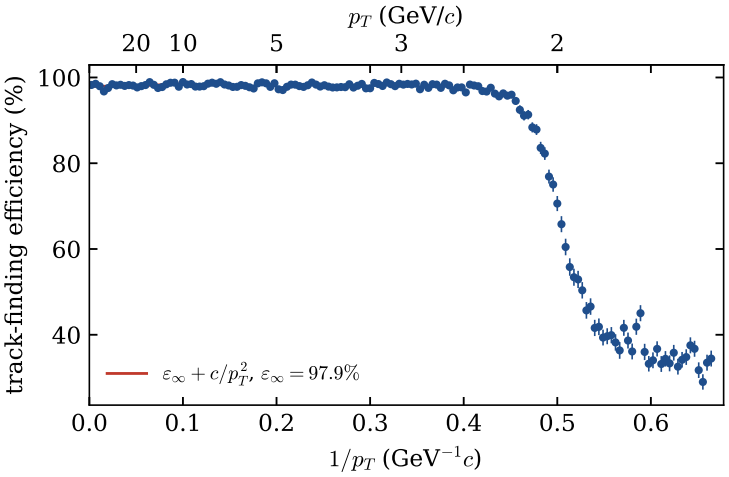


Figure 5: Track-finding efficiency (at least five surviving IT and OT hits) against $1/p_T$, with the fit $\varepsilon_\infty + c/p_T^2$ to the muons above 10 GeV/c.

5 Fake tracks after the selection

The limit. Paper [8] shows that the expected number of combinations of background hits that pass the goodness-of-fit cut of a track fit, with hits on at least k of the layers, is a lower limit on the number of fake tracks of any reconstruction algorithm that keeps the efficiency for real tracks given by that cut. For layers of any shape and noise densities ρ_i that vary across them,

$$E(\geq k) = \int \mathrm{d}^4\theta \sum_{\substack{S\subseteq C(\theta)\\ |S|\geq k}} V_{n_\mathrm{dof}}\, c_S^{n_\mathrm{dof}/2} \sqrt{\det\left(J_S^\mathsf{T} J_S\right)} \prod_{i\in S} \rho_i\, \sigma^2, \tag{3}$$

where θ are the helix parameters, $C(\theta)$ the layers crossed, J_S the derivatives of the crossing points on the layers of S in units of the resolution, c_S the χ^2 cut and V_d the volume of the unit d-sphere. The layers are the six barrel cylinders and the eleven disks of each endcap, with their actual extents. We require at least five layers, with a one-sided 3σ cut for each set of layers. The integral is evaluated by sampling helices with p_T above 10 GeV/c and polar angles in both hemispheres, with the densities after the cuts taken from the background sample as functions of $|z|$ on each barrel layer and of r on each disk; the azimuthal symmetry of the BIB lets us sample a narrow range of azimuth.

Three regions. Each combination is assigned to a region by the layers it uses: the *barrel* if all its layers are barrel cylinders, the *endcap* if all are disks, and the *intermediate* region if it uses both. Every combination belongs to exactly one region, and the three add up to the total. Figure 6 illustrates what a combination looks like in each region: the pieces of the layers crossed by a narrow bundle of tracks, with one track and a few background hits.

The acceptance. The limit refers to a given acceptance: the tracks that the algorithm must find. Real tracks come from the luminous region, and restricting the longitudinal origin of the fitted track removes the fakes that point elsewhere at no cost in efficiency. We take a luminous region of 1.5 mm, the bunch length foreseen at a 10 TeV collider [2], and require $|z_0| < 4.5$ mm for the fitted track. This *track* z_0 should not be confused with the per-hit z_0 of section 3, whose cut is tens of millimetres wide: the fit of five or six hits measures the origin of a real track to 0.1–0.3 mm. Without it the IT and OT would have 1.4 expected fakes per bunch crossing; with it, 76 times fewer.

Results. Table 3 gives the expected fakes in the three regions. With $|z_0| < 4.5$ mm the IT and OT together have

$$E(\geq 5) = 1.9 \times 10^{-2} \text{ fakes per bunch crossing}, \tag{4}$$

almost entirely from five-layer combinations: the six-layer term is 3×10^{-6}. The barrel contributes 90%, the intermediate region 9%, and the endcaps a negligible 10^{-4} of the total. Figure 7 shows how the fakes are distributed in polar angle. In the barrel they concentrate at $45° < \vartheta < 76°$ (both hemispheres folded together): the background density after the cuts is highest at intermediate $|z|$ in every layer, up to 1.4–1.8 times its mean, and these tracks cross the layers there, obliquely. The intermediate region covers 30°–53° and the endcaps 13°–34°. The endcaps are so quiet because the densities on the disks after the cuts are low, 10^{-4}–10^{-3} mm^{-2} except on the first IT disk.

For comparison, one projective tower of six square planes with an outermost plane of 1 m^2, placed at $\vartheta = 90°$, gives 4.1×10^{-3}: the simplest example of ref. [8], with the measured densities. Since the five-layer terms dominate, the expected number grows as the fifth power of a common scale factor of the densities; one fake per crossing is reached at 2.2 times the simulated density (figure 8). The number of distinct fakes is smaller than E by the factor $e^{-\eta/2}$, where the window

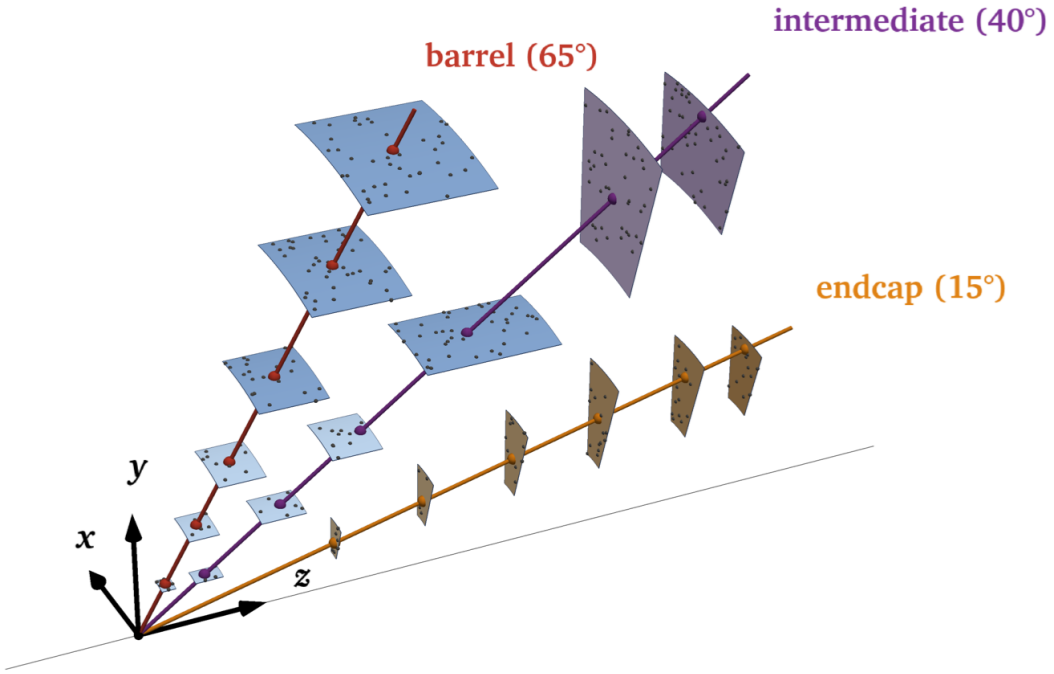


Figure 6: One "tower" per region at its true position: the pieces of the actual layers crossed by a bundle of tracks from the interaction point within ±6° in polar angle and ±9° in azimuth, around a track (with its hits) at $\vartheta = 65°$ (barrel: six barrel layers), 40° (intermediate: four barrel layers and two OT disks) and 15° (endcap: six IT disks). Colours as in figure 1. The small grey spheres are background hits, drawn for illustration only: their number is proportional to the area of each piece, not to the actual density. The expected numbers of fakes of table 3 are for the whole regions, not for these towers.

Table 3: Expected fakes per bunch crossing in the IT and OT (both endcaps, full azimuth), at least five layers, by region, with and without the restriction on the longitudinal origin of the fitted track. The statistical uncertainties of the integration are below 1%.

region	$\lvert z_0\rvert < 4.5$ mm	no z_0 restriction	polar angle (10–90%)
barrel	1.7×10^{-2}	1.1	45°–76°
intermediate	1.7×10^{-3}	0.27	30°–53°
endcap	1.5×10^{-6}	2.7×10^{-4}	13°–34°
total	1.9×10^{-2}	1.4	

occupancy η measures how often accepted combinations share hits [8]; we expect $\eta \ll 1$ at these densities [compute η for the three regions].

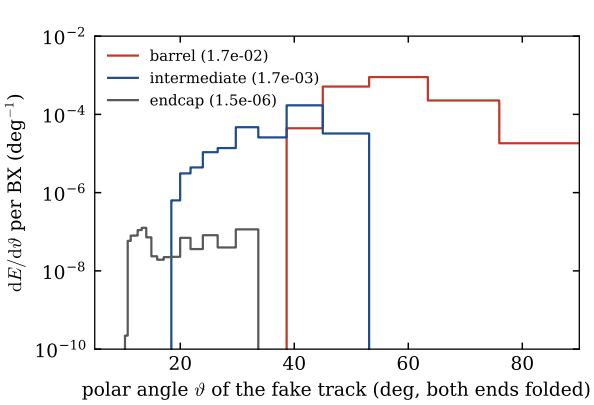


Figure 7: Distribution of the expected fakes in polar angle, for the three regions, with $|z_0| < 4.5$ mm (both hemispheres together). The binning follows equal steps in $\cot\vartheta$.

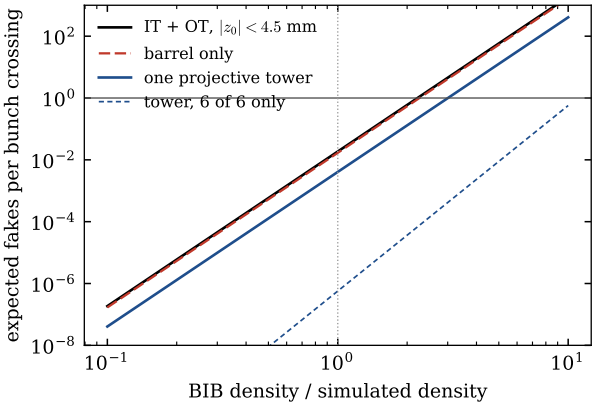


Figure 8: Expected fakes per bunch crossing against a common scale factor of the background densities: IT and OT, barrel alone, and one projective tower.

Margin. A factor 2.2 in density is the margin against an underestimate of the BIB in the simulation, against statistical fluctuations of a single simulated bunch crossing, and against everything the limit leaves out (section 7). Equivalently, at the simulated density the limit stays below one per crossing for resolutions up to about twice the assumed ones (section 6).

What the cuts buy. The same limit, computed with the background densities before the cuts or after only part of them, shows how much each per-hit measurement contributes (figure 9, table 4). Each cut alone reduces the mean densities by a factor between 4 and 25, depending on the subsystem; the two together by 37 to 62. Since the expected number of fakes grows as the fifth power of the densities, these factors are amplified enormously. Without any cut the formula gives 7×10^6 accepted combinations per bunch crossing: at that level the number is no longer a meaningful count of distinct fakes (most combinations share hits, $\eta \gg 1$), but it shows that track finding with the hit positions alone would be hopeless. The time cut alone, or the z_0 cut alone, still leaves 10^2–10^3 combinations per crossing, almost all in the barrel. Only the two together bring the limit below one, to 5.9×10^{-2}, and the momentum cut of the OT, which uses the azimuthal component of the same direction measurement, gains a further factor of three. Neither a time nor a direction measurement is enough by itself: it is their combination on every hit that makes the background manageable.

Table 4: Expected fakes per bunch crossing ($|z_0| < 4.5$ mm, at least five layers) with the background densities before the cuts and after part of them. The last row is the selection of section 4 (table 3). Above about one, the numbers count accepted combinations, many of which share hits, rather than distinct fakes.

cuts applied	barrel	intermediate	endcap	total
none	6.9×10^6	4.7×10^5	3.0×10^3	7.4×10^6
time only	104	0.62	7.9×10^{-4}	104
z_0 only	632	5.4	0.36	637
time and z_0	5.4×10^{-2}	4.6×10^{-3}	4.1×10^{-6}	5.9×10^{-2}
time, z_0 and p_T (OT)	1.7×10^{-2}	1.7×10^{-3}	1.5×10^{-6}	1.9×10^{-2}

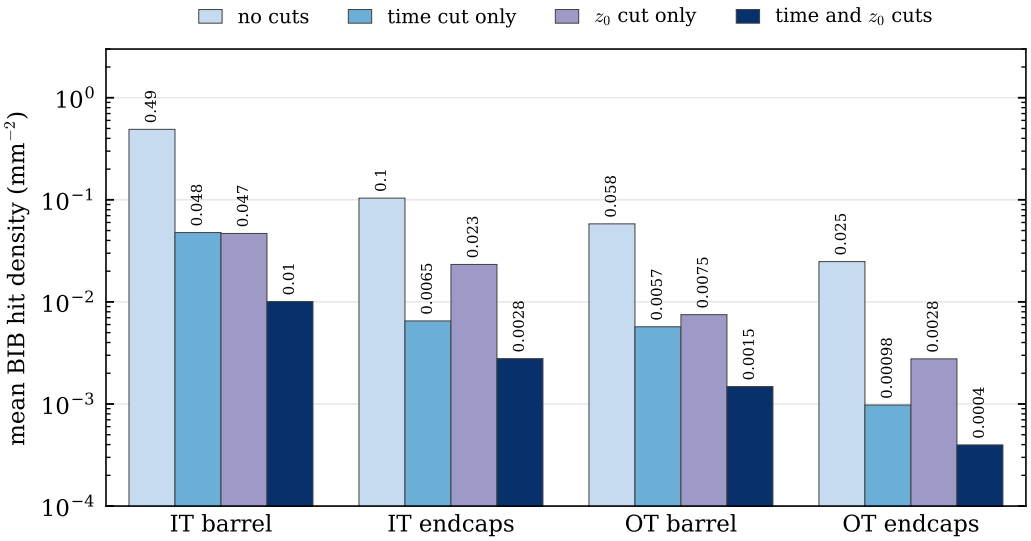


Figure 9: Mean BIB hit density per bunch crossing in the four parts of the IT and OT with no cuts, with the time cut only, with the z_0 cut only, and with the two combined (cuts of table 2). The momentum cut of the OT, not included here, lowers the last values in the OT barrel and endcaps to 8.1×10^{-4} and 2.6×10^{-4} mm^{-2}.

6 Resolution requirements

Scans. We vary one resolution at a time around the operating point (100 μm, 100 ps, 1.5°), from ten times better to ten times worse, keeping the other two fixed. At every point the time and z_0 cuts are derived again at 98% signal containment, so the efficiency stays constant by construction, and the background densities after the cuts and the fake-track limit are recomputed. The scans are done for the barrel, which contains 90% of the fakes. Figure 10 shows the limit for one tower and for the whole barrel, scaled from the tower by their ratio at the operating point (4.1), and the track-finding efficiency. The level of one fake per crossing in the whole barrel is reached at a position resolution of 200 μm, a time resolution of 220 ps or a direction resolution of 4.1°.

Why the curves differ. The position resolution enters twice: through the per-hit observables, whose cuts barely change, and through the χ^2 cut of the track fit, which makes the probability of a five-layer combination scale as σ^6 [8]. The limit therefore falls by six orders of magnitude for a ten-fold improvement, steadily over the whole range. The time and direction resolutions enter only through the cuts, by changing the density of background hits that survive them, raised to the fifth power. The time scan is steep around the operating point, where the time window cuts into the late peak of the background; at a few hundred picoseconds the window contains the whole peak and the curve flattens, the remaining rejection coming from the other two cuts. Below 30 ps the improvement continues as the window closes on the prompt tail of the background. The direction scan is gentler at small angles, where the z_0 window is no longer set by the resolution but by the multiple scattering of the signal muons (10 mm in the IT and 83 mm in the OT at 0.1°), and steepens above 3°; above 6° the z_0 window of the OT becomes so wide that delayed hits of low-momentum muons curling in the field enter the time-cut distribution, the OT time window opens to several nanoseconds and the limit jumps by two orders of magnitude.

Targets for R&D. Table 5 gives the resolution at which the whole barrel reaches given levels of the limit, each resolution varied alone. Present-day values of position and time resolution,

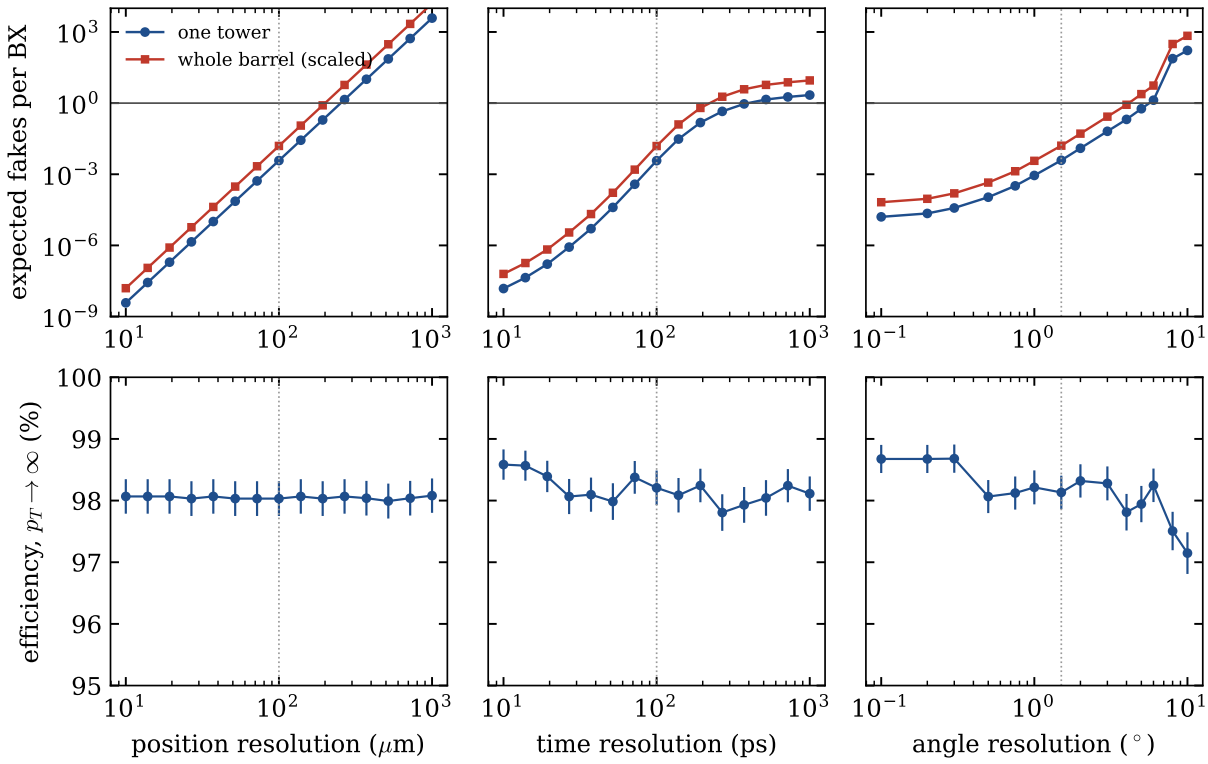


Figure 10: Top: expected fakes per bunch crossing against the position, time and direction resolutions, varied one at a time around the operating point (dotted line), for one projective tower (blue) and for the whole barrel with $|z_0| < 4.5$ mm (red, scaled from the tower by their ratio at the operating point). Bottom: track-finding efficiency for $p_T \to \infty$ of muons confined to the barrel.

100 μm and 100 ps, leave a margin of about two in background density; reaching 10^{-4} fakes per crossing, a margin of about six, requires about 45 μm or 50 ps. A direction measurement better than about 1.3° is needed for the limit to stay below 10^{-2}; it is the least developed of the three capabilities and the one on which the selection depends most when it is degraded.

Table 5: Resolution at which the expected fakes per bunch crossing in the whole barrel ($|z_0| <$ 4.5 mm, at least five of six layers) reach the given level, with the other two resolutions at their operating values (100 μm, 100 ps, 1.5°).

expected fakes per crossing	1	10^{-2}	10^{-3}	10^{-4}
position (μm)	200	93	63	43
time (ps)	220	94	67	48
direction (deg)	4.1	1.3	0.67	0.21

7 Discussion and limitations

The background sample. The results rest on a single simulated bunch crossing. The densities after the cuts are measured from 15 000 to 41 000 hits per layer. Their Poisson precision is below 1%, but hits produced by the same background particle are correlated, and the limit, through its fifth-power dependence, multiplies any relative error on the densities by five. Independent smearings of the same sample change the tower limit by 8% (from 4.1×10^{-3} to 3.8×10^{-3}). The μ^+ and μ^- samples give consistent densities. [bootstrap over BIB primaries for a proper uncertainty]

What the limit does not include. The limit counts combinations made only of background hits, with Gaussian resolution and no multiple scattering, for tracks above 10 GeV/c from the luminous region. Clusters of several hits from one particle, merged hits, sensor inefficiency and noise, combinations of background with signal hits and correlated background such as low-momentum particles curling in the field can only add fakes or lose real tracks. A real algorithm will produce more fakes than the limit; the limit says how many are unavoidable.

Scans of the other regions. The resolution scans cover the barrel only. The intermediate region, with a tenth of the fakes, would cross one fake per crossing at somewhat worse resolutions; in the endcaps the per-hit z_0 is weak and the margin rests on the time cut and on the low densities, so the time resolution matters most there. [scans per region]

Cut optimization. The cuts are set by signal containment alone. Optimizing them against the background, or combining the three observables into one likelihood, may gain further, at the same efficiency.

8 Conclusions

With a time and a direction measured on every hit, three cuts that test the compatibility of each hit with a particle from the interaction point reduce the beam-induced background in the inner and outer trackers of a 10 TeV muon collider by a factor of 57, keeping 97% of the signal hits and 98% of the tracks. With present-day position and time resolutions, 100 μm and 100 ps, and a direction resolution of 1.5°, the algorithm-independent lower limit on fake tracks is 1.9×10^{-2} per bunch crossing, 90% of it in the barrel, 9% in the intermediate region and almost nothing in the endcaps; this is a factor 2.2 in background density below one fake per crossing: the background

does not make track reconstruction impossible in principle. Without the cuts the same limit is 7×10^6, and with either the time or the z_0 cut alone it is still 10^2–10^3: both measurements are needed. The margin grows rapidly with better resolutions, by six orders of magnitude for a ten-fold improvement in position resolution, and the scans give the resolutions needed for any chosen margin. The direction measurement is the most demanding of the three requirements and a natural target for detector R&D.

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A Efficiency for $p_T \to \infty$

The hit and track efficiencies are quoted for $p_T \to \infty$, the limit relevant for the high-momentum tracks to which the fake limit refers. For muons above 10 GeV/c the efficiency is fitted with $\varepsilon(p_T) = \varepsilon_\infty + c/p_T^2$, the form expected from multiple scattering and from the p_T-dependent curvature, whose effects on the observables scale as $1/p_T^2$ in the variance. The fit describes the data well (figure 5); for the track efficiency it gives $\varepsilon_\infty = 97.9 \pm 0.2\%$ and $c = 0.60$ GeV^2. [from docs/efficiency_pt_infinity.pdf: justification and checks]

B Code and data availability

The analysis code, the settings files of the runs used and the scripts that make the figures are available in a public repository [9]. [The input samples: on request, or a public location.]

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